KLOUDSPOT | TECHNICAL ASSESSMENT

Project 2: Multi-Tenant Helpdesk & Ticketing System

Position	Senior Software Engineer (Experienced)
Assessment duration	24 hours from receipt of assessment
Technical point of contact	Dileep | dileep.kotte@kloudspot.com
Mandatory HR recipient	hr.team@kloudspot.com

1. Objective

Build a web application where multiple organizations can manage support tickets. Each organization must have its own users and tickets, and users must not be able to access another organization's data.

2. Functional requirements

User management

Implement login and support roles: Organization Admin, Support Agent, and Customer.

Organization management

Support multiple organizations and associate users and tickets with the correct organization.

Ticket management

Allow users to create, view, update, assign, and close tickets. Each ticket must contain subject, description, priority, status, requester, assigned agent, and timestamps.

Search and filters

Support searching and filtering tickets by status, priority, and assigned agent, with pagination.

Comments and history

Allow comments and maintain a basic history of status and assignment changes. Internal agent notes must not be visible to customers.

REST APIs

Provide appropriate APIs with request validation, proper HTTP status codes, and error handling.

Frontend

Develop a simple interface for login, ticket listing, ticket creation, ticket details, and ticket management.

Database

Persist data using MongoDB, MySQL, or PostgreSQL.

3. Technology stack

- Java 17 or later.
- Spring Boot and Spring Security.
- Spring Data MongoDB or Spring Data JPA.
- Any frontend technology.
- JUnit 5 and Mockito.

4. Deliverables

- Complete source code in a Git repository or ZIP file.
- README with setup and execution instructions.
- Swagger/OpenAPI documentation or a Postman collection.
- Unit tests for important business logic.

5. Evaluation criteria

Java and Spring Boot fundamentals, REST API design, database modeling, authentication and authorization, tenant-level data isolation, code quality, and testing.

6. Submission instructions

- Complete and submit the assessment within 24 hours of receiving it.
- Reply to the assessment email with your GitHub repository link or attach a ZIP file containing the source code and required deliverables.
- Address your submission to dileep.kotte@kloudspot.com and mandatorily include hr.team@kloudspot.com in the recipients (To or CC).
- Ensure the application runs locally by following the README without requiring manual code changes.

Clearly document any incomplete features or assumptions in the README.

Please ensure the implementation is complete, properly organized, and accompanied by the required documentation.